

1. CAUSALITY — does X actually cause Y?

Individual effect: $\tau_i = Y_i(1) - Y_i(0)$

ATE (average treatment effect) = average of τ_i over **everyone**

ATT (effect on the treated) = average of τ_i over **treated people only**

The key equation:
Observed gap = ATT + Selection bias
Observed gap = $E[Y(1) \mid D=1] - E[Y(0) \mid D=0]$
Selection bias = $E[Y(0) \mid D=1] - E[Y(0) \mid D=0]$
In words: selection bias asks: were the treated people already different, even without treatment?

Steps for a potential-outcomes table

- 1. τ for each row = $Y(1) - Y(0)$
- 2. ATE = average τ of all rows
- 3. Observed gap = avg $Y(1)$ of treated rows - avg $Y(0)$ of untreated rows
- 4. ATT = average τ of treated rows
- 5. Selection = avg $Y(0)$ of treated - avg $Y(0)$ of untreated
- 6. Check: ATT + Selection = Observed gap

Big ideas

- Fundamental problem of causal inference: we only ever see ONE outcome per person (treated or not). The other one — what would have happened — is the **counterfactual**, and it's missing.
- Randomized experiment (RCT): chance decides who is treated, so the groups are alike on average → selection bias = 0 → (treatment avg - control avg) estimates the **ATE**.
- RCT problems: dropouts/non-compliance, cost, ethics, may not apply elsewhere (**external validity**).
- Prediction ≠ causation: X can predict Y because it tracks hidden factors; a policy that changes X doesn't change those factors, so Y may not move.
- Correlation ≠ causation. Other reasons X and Y move together: Y causes X (**reverse causality**); a third factor drives both (**confounder**); both just trend over time; pure chance.
- Deterministic: Y is known exactly from X (*F = 1.8°C + 32).
Stochastic: $Y = \beta_0 + \beta_1 X + u$ — there's a random part u .

Four types of data

Type	Who / when	Example
Cross section	many units, one time	survey of workers in 2024
Time series	one unit, many times	U.S. GDP every quarter
Repeated cross sections	new sample each period	different students surveyed each fall
Panel	same units every period	same students re-surveyed each fall

Only a **panel** shows how the same person/district changes over time.

2. PROBABILITY & STATISTICS

PMF = the table of $P(X = x)$. Valid if every value is ≥ 0 and they **add to 1**.

CDF $F(x) = P(X \leq x)$ = running total. Looks like a **staircase**: flat between values, jumps up at each value.
 $P(X > x) = 1 - F(x)$ $P(X \geq 2) = 1 - P(X \leq 1)$

Mean: $E[X] = \sum x \cdot P(X = x)$
Variance: $\text{Var}(X) = E[X^2] - (E[X])^2$, with $E[X^2] = \sum x^2 \cdot P(X = x)$
SD = $\sqrt{\text{Var}}$ — same units as X (variance is in units²)
In words: mean = probability-weighted average; SD = typical distance from the mean.

If $C = a + bX$ (e.g. $C = 30 + 8X$):
 $E[C] = a + bE[X]$ $\text{Var}(C) = b^2 \text{Var}(X)$ $\text{SD}(C) = |b| \text{SD}(X)$
In words: adding a constant moves the mean but not the spread.

Sample statistics (from a list of numbers)

$\bar{x} = \frac{\text{sum}}{n}$ $s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$ **— divide by $n - 1$, not n**

Steps for s^2 : find the mean → subtract it from each value → square each → add up → divide by $n - 1$. Then $s = \sqrt{s^2}$.

Median = middle value once sorted. **Mode** = most common.
Change units (multiply every value by c , e.g. minutes → seconds, $c = 60$): mean, median, SD all $\times c$; variance $\times c^2$.

Covariance & correlation
 $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$ + means they move together, - means opposite.

$\text{Corr} = \rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$ always between -1 and 1, no units.

Corr = 0 only means no straight-line relationship. Example: $Y = X^2$ with X centered at 0 has Corr = 0 but Y depends completely on X .

Conditional probability
 $P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)}$
In words: “out of the B cases, what share are also A ?” — B is the reference group. In a 2×2 table: “ A and B ” = inside cell, $P(B)$ = its total.
Independent if $P(A \text{ and } B) = P(A) \cdot P(B)$ (same as $P(A \mid B) = P(A)$)
If A is inside B (e.g. $X \geq 2$ is inside $X \geq 1$): $P(A \text{ and } B) = P(A)$, so $P(X \geq 2 \mid X \geq 1) = \frac{P(X \geq 2)}{P(X \geq 1)}$.

0/1 variables R and L: $E[RL] = P(\text{both} = 1)$, so $\text{Cov} = P(\text{both}) - P(R)P(L)$. Positive Cov still doesn't prove causation.

3. THE NORMAL DISTRIBUTION

$N(20, 5^2)$ means SD = 5 (the 2nd number is the VARIANCE).

Step 1: z-score (# of SDs from mean): $z = \frac{x - \mu}{\sigma}$

Step 2: look up $\Phi(z) = P(Z \leq z)$ (“phi”) in the table and use:
 $P(X \leq b) = \Phi(z_b)$ $P(X > b) = 1 - \Phi(z_b)$
 $P(a \leq X \leq b) = \Phi(z_b) - \Phi(z_a)$ $\Phi(-z) = 1 - \Phi(z)$

68–95–99.7 rule: about 68% within 1 SD of the mean, 95% within 2 SD, 99.7% within 3 SD. 95% range $\approx \mu \pm 2\sigma$.

Why $P(X = 20) = 0$: for a continuous variable, probability = area under the curve. A single point has no width → zero area. A range like 19 to 21 has width → positive area.

z	.0	.1	.2	.3	.4	.5	.6	.7	.8	.9
0.	.500	.540	.579	.618	.655	.691	.726	.758	.788	.816
1.	.841	.864	.885	.903	.919	.933	.945	.955	.964	.971
2.	.977	.982	.986	.989	.992	.994	.995	.997	.997	.998

$\Phi(1.3)$: row “1.”, column “.3” → .903. $\Phi(1.96) = .975$

4. ESTIMATORS — how good is a guess?

Estimator = the formula/rule (random — changes from sample to sample). **Estimate** = the actual number you get from your data.

Bias = $E[\hat{\theta}] - \theta$. **Unbiased** = right on average.

Efficient: among unbiased ones, the one with **smaller variance**.

Consistent: gets closer to the truth as n grows (bias → 0 and variance → 0).

MSE = Variance + Bias² (lower is better)

In words: a little bias can be worth it if the variance drops a lot.

Tools: $E[\bar{X}] = \mu$, $\text{Var}(\bar{X}) = \sigma^2/n$. For 0/1 data: $E[S] = np$, $\text{Var}(S) = np(1 - p)$.

Any rule $\frac{aS+b}{c}$: $E = \frac{a np + b}{c}$, $\text{Var} = \frac{a^2 np(1-p)}{c^2}$, Bias = $E - p$. (Same idea for $\bar{X}$: $E[k\bar{X}] = k\mu$, $\text{Var} = k^2 \sigma^2/n$.)

Estimator	Bias	Variance	Consistent?
$\hat{p} = S/n$	0	$p(1 - p)/n$	Yes
$\tilde{p} = \frac{S+1}{n+2}$	$\frac{1-2p}{n+2}$	$\frac{np(1-p)}{(n+2)^2}$	Yes
X_1 (for p or μ)	0	$p(1 - p)$ or σ^2	No — ignores extra data
$\bar{X}$	0	σ^2/n	Yes
$\frac{n}{n+1} \bar{X}$	$\frac{-\mu}{n+1}$	$\frac{n\sigma^2}{(n+1)^2}$	Yes

$\bar{X}$ beats X_1 (smaller variance). Median of 0/1 data only tells you whether most answers are 1 — it doesn't estimate p .

5. REGRESSION (OLS)

Model: $Y = \beta_0 + \beta_1 X + u$. **OLS** picks the line that makes the sum of squared residuals as small as possible.

$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$ $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$

Also $\hat{\beta}_1 = r \cdot \frac{s_Y}{s_X}$ → slope has the **same sign as the correlation**.

Steps by hand

- 1. Find $\bar{x}$ and $\bar{y}$.
- 2. Make columns: $x - \bar{x}$, $y - \bar{y}$, $(x - \bar{x})^2$, $(x - \bar{x})(y - \bar{y})$.
- 3. Add up the last two columns → plug into $\hat{\beta}_1$, then $\hat{\beta}_0$.
- 4. Predicted: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$. Residual: $\hat{u} = y - \hat{y}$.
- 5. Check: residuals add to 0, and $\sum x\hat{u} = 0$.

Where formulas come from: minimize $Q = \sum (y_i - b_0 - b_1 x_i)^2$ (squared so + and - misses don't cancel). Set derivatives to 0:
 $\frac{\partial Q}{\partial b_0} = -2 \sum (y_i - b_0 - b_1 x_i) = 0 \rightarrow \sum \hat{u}_i = 0$
 $\frac{\partial Q}{\partial b_1} = -2 \sum x_i (y_i - b_0 - b_1 x_i) = 0 \rightarrow \sum x_i \hat{u}_i = 0$
Divide the first by $\bar{n}$: $\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$ → the line always passes through $(\bar{x}, \bar{y})$.

Changing units: multiply y by c → slope and intercept both $\times c$. Multiply x by c → slope $\div c$, intercept same. (Dollars → cents: slope $\times 100$.) Why: the top of the $\hat{\beta}_1$ formula scales by $c_x c_y$, the bottom by c_x^2 , then $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$. Same fitted line, just new units.

How to interpret (fill in the blanks)

Slope: “Each additional [unit of X] is associated with a $\hat{\beta}_1$ [units of Y] change in Y, on average.”

Intercept: predicted Y when X = 0 (often meaningless if X = 0 is outside the data).

Prediction: plug x into $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$. If x is outside the data range → **extrapolation**, less reliable.

Not causal because: units with more X may differ in other ways (ability, income) that sit in u ; no random assignment.

Key assumption: $E[u \mid X] = 0$

In words: the “other factors” average out to zero at every value of X .

If true, the line gives the **average Y at each X**. You don't need it just to calculate $\hat{\beta}$ — only to interpret it. It fails if, e.g., family income affects test scores AND is related to class size.

Residuals always adding to 0 is just math — it does not prove $E[u \mid X] = 0$.

Reading R output: the “Estimate” column gives the line, e.g. Score = 698.93 - 2.28 · ClassSize. Degrees of freedom = $n - 2$, so 418 df → $n = 420$. R^2 (e.g. .051) = share of the variation in Y explained by the line (5.1%). Descriptive only — not causal.

6. GLOSSARY — every symbol

Greek letters			
$\beta_0, \hat{\beta}_1$	true intercept, slope	$\hat{\beta}_0, \hat{\beta}_1$	estimated int., slope
μ (mu)	population mean	σ, σ^2	pop. SD, variance
σ_{XY}	pop. covariance	ρ (rho)	pop. correlation
τ_i (tau)	i's treatment effect	$\theta_i, \hat{\theta}$	true value; our guess
$\Phi(z)$ (phi)	$P(Z \leq z)$ (table)	ϵ (epsilon)	tiny positive no.
Ω, ω	all outcomes; one	Σ (sigma)	add them all up
Letters			
X vs. x	random var. vs. value	X_i, x_i	observation no. i
Y, X	outcome; explanatory	$Y_i(1), Y_i(0)$	if treated / not
D_i	1 = treated, 0 = not	u	error: other factors
$\hat{y}, \hat{u}$	predicted; residual	$\bar{X}, \bar{x}$	sample mean
s^2, s	sample variance, SD	s_{XY}, r_{XY}	sample cov., corr.
N, n	pop. size, sample size	$p, \hat{p}, \bar{p}$	true prob.; estimates
S	number of 1s in sample	Z, z	std. normal; z-score
b_0, b_1	candidate int./slope	Q	sum sq. residuals
c, a, b	constants	hat, bar	estimate, average
Functions & operators			
$E[X]$	expected value (avg)	$E[Y \mid X=x]$	avg Y at that X
Var, SD	variance, std. dev.	Cov, Corr	cov., correlation
$P(A)$	probability of A	$P(A \mid B)$	A given B
$A \cap B$	A and B both	$p_X(x)$ PMF	$P(X = x)$
$f_X(x)$ PDF	density height	$F_X(x)$ CDF	$P(X \leq x)$
$\sim$	“distributed as”	$N(\mu, \sigma^2)$	normal (var. σ^2)
Bernoulli(p)	0/1, $P(1) = p$	$1\{\cdot\}$	1 if true, 0 if not
$\perp$	“independent of”	$\xrightarrow{p}$	converges as $n \uparrow$
$\partial Q / \partial b$	derivative	arg min	values that minimize
Abbreviations			
ATE / ATT	effect: all / treated	RCT	randomized trial
MSE	mean squared error	OLS	ordinary least squares
SSR	sum sq. residuals	df	deg. of freedom = $n - 2$
iid	indep., same distrib.	STR	class size
OVB	omitted variable bias	IV / DiD / RD	quasi-expt designs
lm(Y~X)	R: regress Y on X		